

Optimal Monetary Policy Rules for Australia*

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Abstract

Using the Reserve Bank of Australia's macro-econometric model MARTIN and their multi-sector DSGE model we simulate alternative monetary policy rules. We find that the current policy, which sets interest rates in response to both trimmed-mean inflation and the unemployment gap performs well. However targeting the price level in conjunction with the output gap, would further reduce inflation volatility and improve aggregate welfare. We also consider a range of nominal aggregates as policy targets, but we find that they are strictly inferior to a flexible inflation target. Among these nominal aggregates, targeting nominal consumption generates the lowest loss, while targeting nominal gross domestic product produces the highest. These results are robust to an increase in the volatility and correlation of future shocks to the Australian economy that might be caused by worsening climate change.

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1 Introduction

The choice of target for monetary policy is the most consequential decision a central bank must make. Since the 1996 the Reserve Bank of Australia (RBA) has used a flexible inflation target to fulfill its mandate of price stability and full employment. The flexible inflation target aims to keep inflation between 2 and 3 per cent on average, over the business cycle, while also placing some weight on the level of unemployment when setting interest rates. It does not seek to make up for periods in which the inflation rate has been outside the target band - it lets bygones be bygones.

The external Review of the Reserve Bank of the Australia has raised the question of whether the current framework is optimal. In particular the Review raises the possibility that a future economy affected by climate-change induced higher volatility may require a different monetary policy framework to best fulfill of the RBA's legislated charter of a) the stability of the currency of Australia; b) the maintenance of full employment in Australia; and c) the economic prosperity and welfare of the people of Australia.

This paper seeks to answer this question by simulating the impact of alternative monetary policy rules using the RBA's macro-econometric model MARTIN and their Multi-Sector DSGE Model (MSM). We first consider the narrow question of how the flexible inflation target should best be structured to best satisfy the RBA's mandate. Should monetary policy focus on current inflation rate only or should the RBA seek to make up for past misses with some form of average inflation targeting? Or alternatively should the RBA adopt a price level target, in which the cash rate is set to keep the price level close to a set trajectory? We find that conditional on maintaining an inflation target the optimal policy rule consists of a price level target, in which trimmed-mean inflation rate is averaged over the history of the model, and the output gap. However the difference between a price level target and a policy rule that averages the inflation rate over the past three years is quantitatively small.

We next consider a range of nominal aggregates as policy targets that could substitute for the current flexible inflation target including a nominal gross domestic product target (in both level and growth terms), a nominal consumption growth rate target, a nominal domestic final demand growth rate target, a nominal household compensation of employees growth rate target (a measure of the aggregate wage bill). We find that these nominal aggregate targets perform poorly compared to rules targeting a weighted average of trimmed-mean inflation and the output gap.

Finally, we test how robust these results are to an increase in the volatility and correlation of future shocks due to climate change. Neither the optimality of a price level target, nor the relative rankings of the nominal aggregate targets are changed if the volatility of shocks increases in the future or the correlation between them increase as might be expected by the changing climate. Macroeconomic volatility increases

under this counter-factual, however the optimal policy rule remains broadly unchanged.

Our analysis adds to a broad literature evaluating optimal monetary policy rules. These analyses are a common and important tool in guiding monetary policy. Clarida, Gali, Gertler (1999), Woodford (2003a, b), and Eggertsson and Woodford (2003) were foundational in analysing optimal monetary policy in New Keynesian models. More recent work has explored the optimality of price level targets (for a recent survey see Hatcher and Minford, 2014), nominal GDP targets (McKibbin and Panton 2018, Beckworth and Hendrickson 2020) and the precise price index which should be optimally targeted (Eusepi et al 2011). This paper is conceptually most similar to Dorich, Mendes and Zhang (2021), who use the Bank of Canada’s model Totem to estimate the optimal monetary policy rule for Canada.

Research on optimal monetary policy in an Australian context is relatively sparse. Gross and Leigh (2022) analyse Australian monetary policy in an historical context. Collins and Siklos (2004) argue that Australia pursued an optimal inflation target in the 1990s. We seek to add to this literature by analysis the optimal monetary policy rule for future Australian monetary policy.

The remainder of the paper is structured as follows. In section 2, we briefly outline the two models of the Australian economy published by the RBA that will be the workhorse of our analysis. In section 3 we analyse the optimal structure of a flexible inflation target and in section 4 we consider a broader set of possible monetary policy rules that target a range of nominal aggregates. In section 5, we discuss how our results are affected by alternative measures of welfare and future changes in the volatility of the economy. The final section concludes.

2 Models

To estimate the impact of monetary policy in Australia we use two of the RBA’s large scale models of the economy. The macro-econometric model MARTIN first published by Ballantyne et al (2020) and a updated version of the Multi-Sector DSGE Model published by Rees, Hall and Smith (2016). The version of the MSM we use is a slightly modified version of the multi-sector model outlined in Rees, Hall and Smith (2016). Our version includes nominal wage rigidities and imperfect labour substitution between each production sector and calculates the flexible-price output gap. Both models are re-estimated up to Q4 2019.¹

2.1 MARTIN

MARTIN is a large model containing 150 equations and covers the household sector, housing market, trade sector, labour market, commodity sector, financial markets and non-financial private corporations.

¹We also considered the extended multi-sector model outlined in Gibbs, Hambur and Nodari (2019). However we found the flexible price output gap had an implausibly high volatility.

We update the version of MARTIN published as Ballantyne et al. (2020) by re-estimating the model up until the end of 2019 and substituting three data series that are not publicly available with publicly released counterparts as described by Stephan (2019). Specifically, we use US variables for World GDP and export prices, and a G3 average (the average of the United States, Japan and the Eurozone) for interest rates. We also have slightly different outcomes for businesses and government investment due to the lack of public data on net asset transfers between the private and public sectors. Details of the data series used and other minor modifications are outlined further in the Appendix.

As a macro-econometric model MARTIN differs from conventional New Keynesian macroeconomic models. It does not contain any forward-looking terms or measures of expectations, nor is it derived using micro-foundations. Instead, MARTIN combines a large number of individually estimated relationships, calibrated equations and economic identities to model the Australian economy. This framework focuses on replicating the empirical relationships in the macroeconomy and combining them in a way that is internally consistent. This modeling approach involves important trade-offs. MARTIN is able to focus on an empirically-focused picture of the economy, potentially capturing aspects that cannot be explained with a micro-founded model. This approach comes at the cost of less clarity on the causal mechanisms at play, and the inability to change deep structural parameters within the model. MARTIN does not explicitly model household and firms' expectations which is a significant constraint when modeling alternative monetary policy frameworks, however this restrictive assumption is relaxed in the MSM.

MARTIN models the impact of monetary policy through several channels of transmission. A lower nominal cash rate increases household consumption via a wealth effect from both housing and financial assets, and via an income effect on both labour and non-labour income. It also boosts investment by increasing the demand for housing and lowering the cost of capital for firms. Finally, a decrease in the cash rate will depreciate the exchange rate thus increasing net exports. These channels all affect aggregate output which drives changes in the unemployment rate. The inflation rate is in turn driven by changes in the unemployment gap according to MARTIN's Phillips curve.

MARTIN's baseline policy rule in sets the nominal cash rate in response to the year-ended trimmed-mean inflation rate and both the level and change in the unemployment gap. To allow for alternative monetary policy targets we change the rule for the nominal cash rate, NCR_t , in MARTIN (equation 1.10.1) with the augmented policy rule

$$NCR_t = 0.7 \times NCR_{t-1} + 0.3 \times \left(r_t^* + \sum_x \mu_x (x_t - \bar{x}_t) \right)$$

Where x_t is a range of alternative policy targets, $\bar{x}$ is the respective target and r_t^* is the real natural interest

rate. We assume that the degree of interest rate smoothing is unchanged from the baseline calibration, while allowing the weights on the target variables to be optimised for each alternative rule.

To approximate welfare across the different counterfactuals, and to select the parameters in the policy rule, we assume a quadratic loss function consisting of deviations of trimmed-mean inflation from the target rate of 2.5 per cent, the unemployment rate from the non-accelerating inflation rate of unemployment (NAIRU), and quarterly changes in the nominal interest rate:

$$L_t = (\pi_t - 2.5)^2 + \lambda^u (u_t - u_t^*)^2 + \lambda^i (i_t - i_{t-1})^2$$

where π_t is the year-ended trimmed-mean inflation rate which is targeted at the midpoint of the RBA's target band of 2 to 3 per cent and $u_t - u_t^*$ is the unemployment gap (the difference between the unemployment rate and the NAIRU) and $i_t - i_{t-1}$ is the quarterly change in the nominal cash rate.

In our baseline results we assume an equal weight on deviations in inflation and unemployment from their respective optimums, while placing a weight of $\frac{1}{2}$ on inflation smoothing following Dorich, Mendes and Zhang (2021) and the RBA's current mandate which places an equal weight on full employment and the inflation rate (Gross and Leigh 2022).

To simulate the economy in MARTIN we draw a set of the estimated reduced form residuals from the model's equations and impose these residuals on the model. When bootstrapping the model residuals we draw all the equations' residuals from a given period to preserve the cross-variable correlation structure of the residuals observed in the data. After each simulation, we calculate the volatilities of the model variables and the quadratic loss function. Repeating this process for a large number of simulations gives us an indication of the unconditional moments of variables in the model. One of the disadvantages of the macro-econometric model approach is that stability (ie avoiding simulations in which the economy spirals off to positive or negative infinity) is neither guaranteed nor straight forwardly achieved. To avoid non-stable simulations we simulated MARTIN over a 10 year sample period only. We find that longer simulation periods tend to increase the share of simulations which are unrealistically unstable. As an additional measure to limit the impact of potentially unstable simulations we trim the top and bottom 2% of all simulations when calculating the macroeconomic volatilities and thus the expected welfare loss.

2.2 Multi-Sector Model

The MSM consists of measures of optimising agents, namely utility maximising households and profit maximising firms. Households derive utility from consumption and disutility from labour services, which

they supply to firms. Households save in bonds and capital, which is specific to each of the three production sectors.

The domestic economy consists of five sectors: a non-tradable sector, a resource sector, a non-resource traceable sector, an imports sector and a sector that produces final goods and services. Firms in the non-tradable, resource and non-resource tradable sectors produce output using labour, capital and resource commodities as inputs. The imports sector purchases goods from overseas and sells them in the domestically. Firms in the non-tradable, non-resource tradable and imports sectors are imperfectly competitive, so that individual firms have some pricing power. By contrast, resource commodities are homogeneous and the price of these goods (in foreign currency) is determined entirely abroad.² The economy exports resources and non-resource tradable goods.

The model's monetary authorities adjust the nominal interest rate to stabilise inflation and aggregate output in level and growth terms. Fiscal policy is specified as an exogenous government spending process that is funded through lump-sum taxation. Nominal rigidities are included in the non-tradable, non-resource tradable and imports sectors in the form of sector-specific quadratic adjustment costs that firms must pay when changing their prices. Wage rigidity, modeled as a quadratic adjustment costs that workers must pay when changing their nominal wage, is included in the non-tradable, non-resource tradable and resource sectors. Economic fluctuations are driven by a number of exogenous shocks including consumption preference shocks, government spending shocks, investment technology shocks, resource price shocks, sector-specific and aggregate technology shocks and mark-up shocks as well as shocks to output, inflation and interest rates in overseas economies.

One limitation of MARTIN is that agent's expectations are not endogenously modeled - they are assumed to be an entirely exogenous variable. The Multi-Sector DSGE Model (MSM) micro-founds firms' and households expectations explicitly. These expectations respond endogenously to the different policy targets being simulated.

In the MSM the nominal interest rate is set in response to the quarterly trimmed-mean inflation rate, the level and growth rate of output and the change in the real exchange rate. Analogous to our approach with MARTIN, we change the monetary policy rule to the more general form of

$$i_t = \rho_r i_{t-1} + (1 - \rho_r) \left(\sum_{x \in X} \rho_x (x_t - \bar{x}) \right)$$

where x_t is a range of potential target variables including the output gap, nominal GDP, trimmed-mean inflation, nominal consumption in both growth and level terms. We assume the central bank follows a

²The MSM abstracts from Australian firms having market power in the commodity sector.

systematic policy rule with no exogenous monetary policy shocks. Following Doric, Mendes, and Zhang (2021) we assume a loss function of the form

$$L_t = \lambda^\pi (\pi_t - 2.5)^2 + \lambda^y (y_t - \bar{y}_t)^2 + \lambda^i (i_t - i_{t-1})^2$$

where π_t is the annual trimmed-mean inflation rate, and $y_t - \bar{y}_t$ is the flexible price output gap and $i_t - i_{t-1}$ is the quarterly change in the nominal cash rate. The weights on the output gap, λ^y , and the trimmed-mean inflation rate, λ^π , are calibrated such that the loss function places an equal weight on the unemployment rate, assuming a Okun's coefficient of 2.³ We assume the weight on interest rate smoothing is $\frac{1}{2}$. These assumptions about the parameterisation of the loss function are relaxed in Section 5.

Because the MSM can be approximated as a linear vector auto-regression derived from the estimated parameters we can calculate the theoretical moments of the model under each policy rule, which abstract from simulation error and initial values. In the baseline results we assume that future exogenous shock processes will be the same as the estimated sample and uncorrelated with each other. We relax this assumption in Section 5 when we consider how climate change may alter the exogenous shocks that effect the Australian economy in the future.

3 Alternative Inflation Targets

The first exercise we conduct is to analysis the optimal structure of the flexible inflation target such that it that best implements the RBA's dual mandate of price stability and full employment. We compare a range of interest rate rules which respond to a combination of the unemployment gap and different measures of inflation. In MARTIN we compare the baseline rule which targets year ended trimmed-mean inflation with alternative rules which respond to the current quarter's inflation rate only and where nominal cash rate is set in response to deviations of the price level from its target trajectory (Table 1).

Table 1: Alternative Inflation Targets in MARTIN

Inflation Measure	Nominal Cash Rate Rules
Average past year	$NCR_t = 0.7 \times NCR_{t-1} + 0.3 \times \left(r_t^* + \mu_\pi \left(\frac{1}{4} \sum_{i=0}^4 \pi_{t-i} - 2.5 \right) + \mu_u (u_t - u_t^*) \right)$
Current quarter	$NCR_t = 0.7 \times NCR_{t-1} + 0.3 \times (r_t^* + \mu_\pi (\pi_t - 2.5) + \mu_u (u_t - u_t^*))$
Price level target	$NCR_t = 0.7 \times NCR_{t-1} + 0.3 \times (r_t^* + \mu_\pi (p_t - \bar{p}_t) + \mu_u (u_t - u_t^*))$

We allow the nominal cash rate's sensitivity to changes in both the inflation measure and the output

³Assuming a normalised weight of $\lambda^\pi = 1$ this implies $\lambda^y = 0.25$.

gap to be optimised in order to minimise the loss function when simulating MARTIN. We find that the two inflation targets performing comparably well, with the current quarter rule having a marginally lower volatility for both inflation and the unemployment gap. By contrast the price level target performs relatively poor with both higher volatility for both the inflation rate, the output gap and the change in the nominal interest rate (Table 2).

Table 2: Volatilities with Alternative Inflation Targets in MARTIN

	Loss	Trimmed- mean Inflation	Unemployment Gap	Change in Nominal Interest Rate
Average past year	1.39	0.45	0.75	0.39
Current quarter	1.36	0.43	0.72	0.41
Price level target	1.80	0.49	0.94	0.75

The forward looking nature of the MSM allows us to model a wider range of alternative inflation targeting frameworks (Table 3). We consider six alternative rules in which the policy rule targets the output gap in conjunction with

1. the expected inflation rate over the next 4 quarters,
2. the expected inflation rate in the next quarter,
3. the inflation rate in the current quarter,
4. the average rate of inflation over the past year,
5. the average rate of inflation over the past 3 years,
6. a price level target, which is the equivalent of averaging the rate of inflation over the entire history.

For each of the alternative policy we choose the parameters ρ_π and ρ_y to minimise the loss function. The smoothing parameter is held fixed at its estimated value of 0.87.

Table 3: Alternative Inflation Targets in the Multi-sector Model

Inflation Measure	Nominal Cash Rate Rules
Expected over the next 4 quarters	$i_t = \rho_r i_{t-1} + (1 - \rho_r) (\rho_\pi E_t \pi_{t+4,t} + \rho_y \tilde{y}_{t,t})$
Expected over the next quarter	$i_t = \rho_r i_{t-1} + (1 - \rho_r) (\rho_\pi E_t \pi_{t+1,t} + \rho_y \tilde{y}_{t,t})$
Current quarter	$i_t = \rho_r i_{t-1} + (1 - \rho_r) (\rho_\pi \pi_t + \rho_y \tilde{y}_{t,t})$
Average past year	$i_t = \rho_r i_{t-1} + (1 - \rho_r) (\rho_\pi \pi_{t,t-4} + \rho_y \tilde{y}_{t,t})$
Average past 3 years	$i_t = \rho_r i_{t-1} + (1 - \rho_r) (\rho_\pi \pi_{t,t-12} + \rho_y \tilde{y}_{t,t})$
Price level target	$i_t = \rho_r i_{t-1} + (1 - \rho_r) (\rho_\pi (p_t - \bar{p}_t) + \rho_y \tilde{y}_{t,t})$

We find that setting the nominal cash rate in response to the deviations of the price level from a fixed trend generates the lowest loss (Table 4). How it does not strictly dominate the other rules; the price level target generates lower inflation volatility and targeting the current or expected inflation rate generates a lower output gap. While this loss minimizing policy rule is backwards looking - in that the cash rate responds to inflation averaged over the past year - it still stabilises current shocks endogenously via household and firms' expectations. For example, a contemporaneous shock today which increases the expected inflation rate over the subsequent year will increase households forecast for the nominal cash rate and thus tighten monetary policy contemporaneously. These results are consistent with Doric, Mendes, and Zhang (2021) who found that a price level target would best stabilise CPI inflation in Canada, at the expense of higher output volatility.⁴

The two models give different results regarding the nature of the inflation target. MARTIN reports a slightly lower loss function when targeting the current quarter's inflation rate, while the MSM favors a price level target, which is the equivalent of targeting the average inflation rate over the entire sample. On balance we believe the MSM's explicit modeling of expectations, a key channel through which monetary policy operates, should lead us to place more weight on its results. Consequently we conclude that the optimal structure of a flexible inflation target would involve a policy rule that responds to the price level's deviation from a target trend path and the output/unemployment gap. However the difference between the backwards looking inflation targets is quantitatively small, so factors that are not well captured by these models may be a deciding factor when determining the optimal monetary policy rule.

For example, a price level target provides a higher degree of accountability and helps stimulate the economy when monetary policy is constrained by the effective lower bound. However it may require periods of distillations when the price level overshoots the target whose costs are not well capture by a linear model which ignores downward nominal wage rigidity. The potentially high cost of disflationary periods is one reason why the Federal Reserve adopted an asymmetric flexible inflation target which only seeks to make up for periods of undershooting the inflation target while ignoring periods of above target inflation once inflation has returned to target (Cleared 2022). This non-linear effect will not be captured in this non-linear analysis and maybe one reason why an asymmetric approach to the inflation target would also be optimal for Australia. A price level target may also pose communication challenges regarding the speed at which the price level will return to it's target.

A more general result we find is that the longer a central bank averages inflation rate over the past the closer you approximate a price level target, with lower inflation volatility, higher output gap volatility and

⁴It is not clear to what extent this would represent a departure from the status quo of policy as while MARTIN's current policy rule uses year-ended trimmed-mean inflation - effectively an average inflation target with a horizon of one year - the MSM only includes the trimmed-mean inflation rate at a quarterly frequency.

a lower loss overall. Thus if a price level target proves too difficult to communicate, it is possible to achieve most of its benefits under a flexible inflation targeting framework that averages the inflation rate over the past 2-3 years instead.

The relatively strong performance of backwards looking inflation targets is in part due to the assumption that households have full information and rational expectations. As Brassil, Gibbs and Ryan (2022) show when households are not rational, but instead set expectations by a backward looking learning process, the optimal policy rule places more weight on expected future inflation.

Table 4: Volatilities with Alternative Inflation Targets in the Multi-sector Model

	Loss	Trimmed-mean Inflation	Output Gap	Change in Nominal Interest Rate
Expected - next 4 quarters	1.34	1.12	0.73	0.08
Expected - next quarter	1.33	1.11	0.73	0.08
Current quarter	1.33	1.09	0.72	0.08
Average past year	1.30	1.05	0.78	0.11
Average past 3 years	1.27	0.99	0.90	0.10
Price level target	1.24	0.94	1.01	0.10

4 Nominal Aggregate Targets

We next consider an set of alternative policy rules in which the nominal cash rate is used to target a single nominal aggregate which combines both a measure of activity as well as the inflation in a single target. We first estimate five alternative policy rules using MARTIN that target nominal GDP in both level and growth terms, and the growth rates of nominal domestic final demand, nominal consumption and nominal compensation of households (a measure of the aggregate nominal wage bill) see Table 5. Each rule is calibrated to minimise the quadratic loss function.

We find that all the nominal aggregate rules perform poorly (Table 6) both in comparison to the baseline policy rule and the alternative inflation targets as reported in Table 4. Among the nominal aggregate rules the nominal consumption target performs least worst, while the nominal GDP targets generates the highest volatility in inflation and the unemployment rate. The poor performance of the latter is due to the inclusion of nominal exports which are highly sensitive to international commodity prices.

A positive commodity price shocks will increase nominal export prices and thus nominal GDP. However because the shock causes the local currency to appreciate it causes trimmed-mean inflation to decrease and

Table 5: Alternative Nominal Aggregate Targets in MARTIN

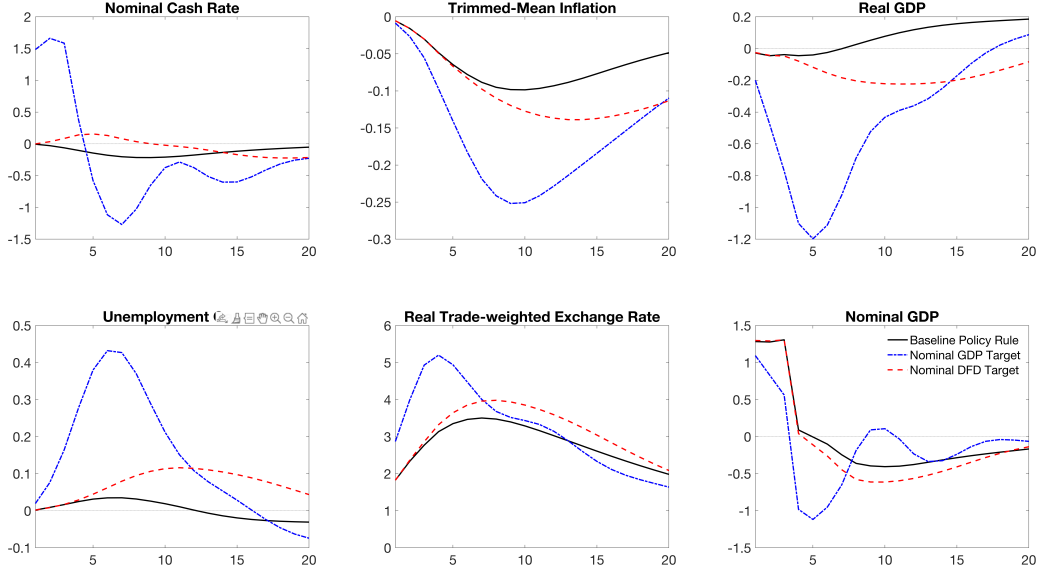
Nominal aggregate	Policy Rule
Nominal GDP Growth Target	$NCR_t = 0.7 \times NCR_{t-1} + 0.3 \times (r_t^* + \mu_{NGDP} (NGPD_t - \overline{NGDP}))$
Nominal GDP Level Target	$NCR_t = 0.7 \times NCR_{t-1} + 0.3 \times (r_t^* + \mu_{NGDP} (NGPDL_t - \overline{NGDPL_t}))$
Nominal DFD Target	$NCR_t = 0.7 \times NCR_{t-1} + 0.3 \times (r_t^* + \mu_{NGDP} (NDFD_t - \overline{NDFD}))$
Nominal HCOE Target	$NCR_t = 0.7 \times NCR_{t-1} + 0.3 \times (r_t^* + \mu_{NHCOE} (NHCOE_t - \overline{NHCOE}))$
Nominal Consumption Target	$NCR_t = 0.7 \times NCR_{t-1} + 0.3 \times (r_t^* + \mu_{NCon} (NCon_t - \overline{NCon}))$

the unemployment rate to rise. A nominal GDP target thus performs poorly in a small commodity exporting economy, exacerbating the impact of commodity price shocks on the domestic labour market and inflation rate. Figure 1 shows the impulse response functions of a commodity price shock in MARTIN under the baseline policy rule, a nominal GDP target and a nominal DFD target. The nominal GDP target generates significantly more volatility in response to the commodity price shock as the policy rule calls for higher interest rates even as inflation and the unemployment gap fall. This effect is mitigated in the nominal aggregates that do not directly include nominal exports, such as domestic final demand, consumption and household compensation. However they still out performed by the targets which respond to a combination of the unemployment gap and inflation.

Table 6: Volatilities with Alternative Nominal Aggregate Targets in MARTIN

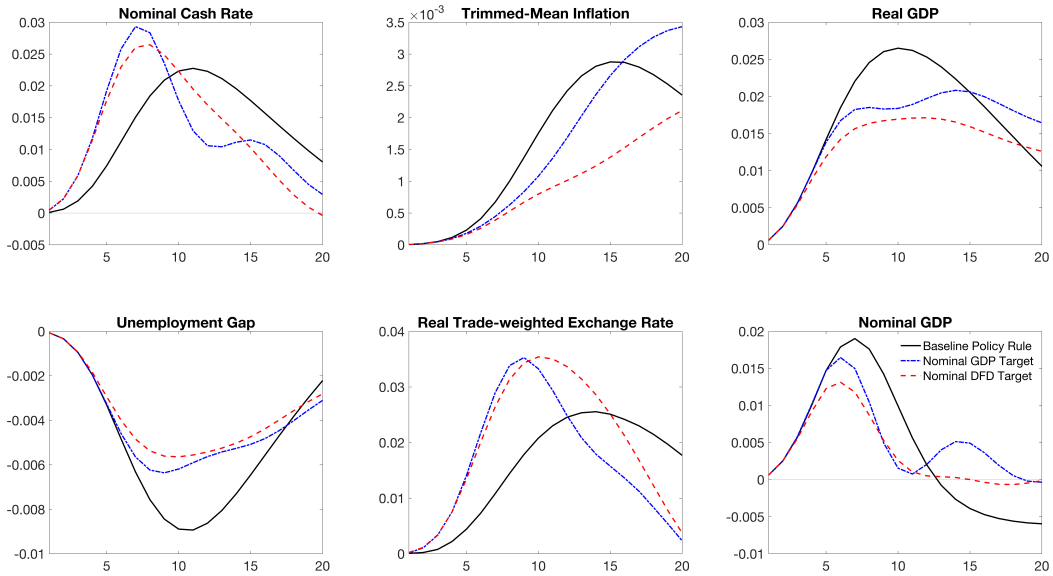
Target	Loss	Trimmed-mean Inflation	Unempl. Gap	NGDP	Change in rates	Nominal Consumption	NDFD	NHCOE
Baseline Policy Rule	1.53	0.55	0.39	8.10	1.19	2.22	4.01	4.25
NGDP Growth	2.19	1.31	0.60	7.11	0.55	1.91	2.72	5.08
NGDP Level	2.74	1.92	0.69	8.77	0.28	3.46	3.59	6.15
Nominal DFD	2.11	1.25	0.58	7.20	0.58	2.09	2.57	7.82
Nominal HCOE	1.93	1.17	0.48	7.23	0.57	1.90	2.98	4.61
Nom. Consumption	1.80	0.98	0.55	7.60	0.55	2.07	3.05	6.27

Figure 1: Commodity Price IRFs in MARTIN



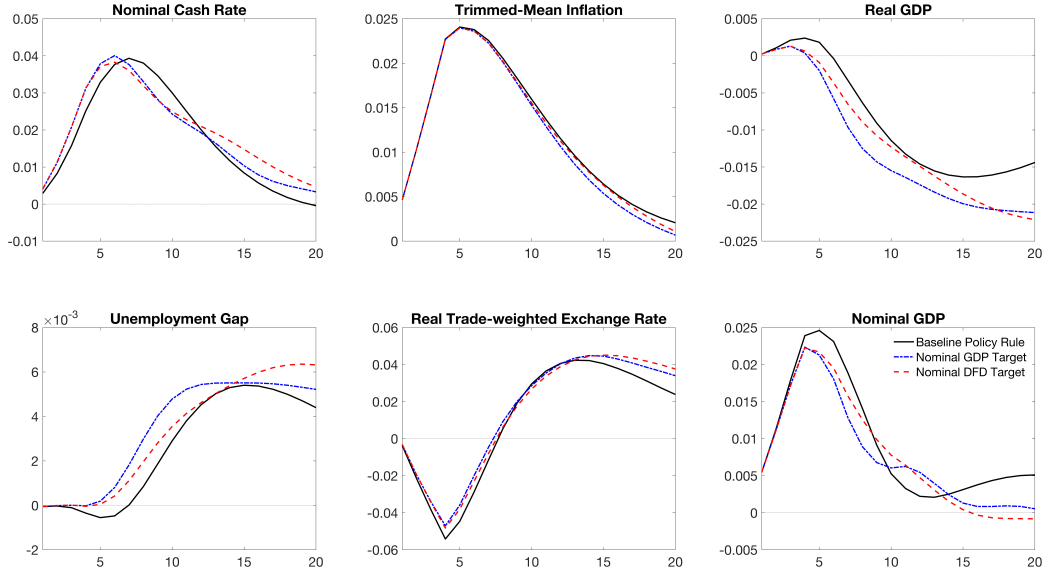
The nominal GDP target also fails to strictly improve on the baseline policy in response to an increase housing demand shock.⁵ As shown in Figure 2 a nominal GDP target leads to higher inflation towards the end of the IRF horizon. Nor does it out perform the baseline policy rule in response to an increase in inflation expectations, which functions similar to a mark up shock in MARTIN (Figure 3) as it produces a larger fall in the level of GDP.

Figure 2: Housing Demand IRFs in MARTIN



⁵ Perhaps driven a by natural disaster which destroys a portion of the housing stock.

Figure 3: Inflation Expectations IRFs in MARTIN



We next simulate two alternative nominal aggregate policy target in the multi-sector model: a nominal GDP target and a nominal consumption target, across the same size timing rules as in the previous section (ie expected growth rates over 4 and 1 quarters, the current growth rate, the average growth rate over the past year and 3 years and a level target).

In the baseline simulation the nominal GDP target fails to generate a stable equilibrium under any timing assumption or parameterisation.⁶

The nominal consumption target does produce a stable equilibrium. However it is strictly dominated by the dual mandate rules which responds to both inflation and the output gap. The volatilities of both inflation and the output gap are significantly higher under a nominal consumption target.

Table 7: Volatilities with a Nominal Consumption Target

	Loss	Trimmed-mean Inflation	Output Gap	Change in Nominal Interest Rate
Expected - next 4 quarters	3.68	1.62	6.41	0.91
Expected - next quarter	3.30	1.54	5.98	0.53
Current quarter	4.75	1.73	10.40	0.84
Average past year	2.47	1.18	4.68	0.24
Average past 3 years	2.33	1.15	4.66	0.03
Level target	2.86	1.11	6.68	0.15

The indeterminate solution for the nominal GDP targets should not be treated as a uninformative result.

⁶The Blanchard-Kahn conditions are not satisfied as monetary policy is not sufficiently reactive to create a single equilibrium path in the model.

A monetary policy rule that does not satisfy the Taylor principle and thus does not stabilise the economy is a very poor policy rule - even if it's cost cannot be quantified. The fact that the model is rendered indetermined is akin to US monetary policy in the pre-Volker era which produced high and volatile inflation as discussed in Bhattarai, Lee and Park (2012). Is the inability of a nominal GDP target to generate a single equilibrium in the DSGE model an example of model misspecification? This result is caused by a slightly negative correlation between the GDP deflator the CPI, which is driven by the commodity price shock which increases nominal GDP via higher nominal exports while decreasing domestic price pressure due to the exchange rate effect. It is conceivable that an alternative model would reverse this result and thus produce a stable result. We also consider a policy rule which combines a nominal GDP target with a trimmed-mean inflation rate target with a fixed weight of 1.5. The inclusion of the latter term guarantees a stable equilibrium within the model. However across all but the three year average target the optimal weight on nominal GDP is zero, and the loss function is strictly dominated by both the nominal consumption rule and the average inflation targeting rule. Given that NGDP targeting performs poorly both when it is combined with an inflation target in the policy rule (to achieve determinacy) and in the the macro-econometric model MARTIN our results suggest that it would be a poor monetary policy rule even ignoring it's inability to stabilise the model.

5 Robustness

Alternative Loss Functions

How robust are these results to the parameterisation of the loss function? In this section we recalculate the optimal policy rules in the MSM under alternative loss functions. Lowering the social planner's aversion to inflation volatility (decreasing λ^π to 0.5) narrows the difference between the policy rules with the price level target producing the same loss as an 3 year average (Table 8). When aversion of inflation is increased (raising λ^π to 2) the ordering of the policy rules is unchanged, with the price level target still having the lowest loss function. Even though a price level target universally generates lower volatility in inflation, at the expense of more volatility in the output gap, a central bank which places half the weight on stabilising trimmed-mean inflation still find it to the optimal policy rule (albeit tied for first place with a three year average inflation target).

This is because the weights on the measure of inflation and the output gap are re-optimised for each rule and alternative loss function. However if the weight on inflation volatility was reduced to zero, then a price

level target would not longer be the optimal policy rule. Removing the interest rate smoothing from the loss function entirely greatly improved the performance of the forward looking policy rules, at the expense of much higher interest rate volatility.

The nominal consumption policy target still performs poorly under all alternative loss functions (Table 9). The best policy rule remains targeting the average nominal consumption growth rate over the past three years. However it is still produces strictly higher inflation and output gap volatility than the flexible inflation target across all alternative loss functions.

Table 8: Optimal Inflation Target with Alternative Loss Functions

		Loss	Trimmed-mean Inflation	Output Gap	Change in Nominal Interest Rate
Lower Inflation Aversion, $\lambda^\pi = 0.5$	Expected - next 4 quarters	0.78	1.13	0.61	0.13
	Expected - next quarter	0.78	1.12	0.61	0.13
	Current quarter	0.79	1.09	0.63	0.13
	Average past year	0.77	1.09	0.61	0.15
	Average past 3 years	0.76	1.05	0.65	0.15
	Price level target	0.76	0.96	0.82	0.15
Higher Inflation Aversion, $\lambda^\pi = 2$	Expected - next 4 quarters	2.47	1.13	0.61	0.13
	Expected - next quarter	2.45	1.11	0.66	0.12
	Current quarter	2.41	1.05	0.97	0.13
	Average past year	2.31	0.96	1.30	0.13
	Average past 3 years	2.24	0.96	1.19	0.06
	Price level target	2.17	0.87	1.31	0.18
No Interest Rate Smoothing, $\lambda^i = 0$	Expected - next 4 quarters	1.05	1.13	0.36	0.79
	Expected - next quarter	1.04	1.13	0.37	0.76
	Current quarter	1.15	1.09	0.40	0.79
	Average past year	1.09	1.08	0.40	0.90
	Average past 3 years	1.06	1.09	0.37	0.85
	Price level target	1.08	0.94	0.66	0.90

A More Volatile Future

One concern with this analysis is that by simulating the models with historical errors based on the sample period (1998 Q4 - 2019 Q4) we are assuming that future volatility in the economy will be similar to the recent historical experience. A changing climate, both in Australia and around the world, might well change the structure of the exogenous shocks the Australian economy is subject to. McKibbin, Morris, Wilcoxon, Panton (2020) discuss how an increased risk of natural disasters such as floods, fires, and droughts will increase the frequency and correlation of exogenous supply shocks. These shocks tend to both increase inflation and reduce output simultaneously posing a challenge to monetary policymakers who can only determine the

Table 9: Nominal Consumption Target with Alternative Loss Functions

		Loss	Trimmed-mean Inflation	Output Gap	Change in Nominal Interest Rate
Lower Inflation Aversion, $\lambda^\pi = 0.5$	Expected - next 4 quarters	2.86	1.65	6.60	0.78
	Expected - next quarter	2.53	1.54	5.98	0.53
	Current quarter	3.88	1.75	10.52	0.75
	Average past year	1.88	1.20	4.69	0.21
	Average past 3 years	1.75	1.16	4.65	0.03
	Level target	2.30	1.10	6.68	0.15
Higher Inflation Aversion, $\lambda^\pi = 2$	Expected - next 4 quarters	5.28	1.58	6.16	1.17
	Expected - next quarter	4.76	1.40	5.23	1.28
	Current quarter	6.48	1.71	10.24	0.98
	Average past year	3.65	1.17	4.68	0.29
	Average past 3 years	3.47	1.13	4.72	0.03
	Level target	3.97	1.11	6.68	0.15
No Interest Rate Smoothing, $\lambda^i = 0$	Expected - next 4 quarters	2.84	1.46	5.52	2.58
	Expected - next quarter	2.71	1.40	5.23	1.28
	Current quarter	4.04	1.57	9.90	6.26
	Average past year	2.33	1.15	4.73	0.37
	Average past 3 years	2.31	1.14	4.68	0.03
	Level target	2.78	1.11	6.68	0.15

response of aggregate demand.

To test our results robustness to higher supply side volatility we first increase the standard errors of the models' shocks when simulating the alternative policy rules. In MARTIN this is achieved by increasing the bootstrapped residuals by 10% across all equations.⁷ In the MSM the standard deviation of the domestic mark up shocks, commodity price shock and world inflation shock only are increased by 10% in an attempt to approximate the impact of a more volatile climate. Table 10 and 11 shows the results of the alternative inflation targets when the shocks' volatilities are increased. While the higher shock volatility decreases welfare under all possible policy rules the relative ordering is not changed.

Table 10: Volatilities with Alternative Inflation Targets in MARTIN with 10% Higher Residuals

	Loss	Trimmed- mean Inflation	Unemployment Gap	Change in Nominal Interest Rate
Average past year	1.69	0.54	0.92	0.47
Current quarter	1.65	0.52	0.88	0.50
Price level target	2.26	0.63	1.18	0.91

The same is true of the nominal aggregate rules (Table 12). Even assuming a higher volatility for

⁷Due to the reduced form nature of the MARTIN's equations it is not possible to selectively increase the volatility of the supply side shocks only.

Table 11: Volatilities with Alternative Inflation Targets in the MSM with 10% Higher Standard Errors

	Loss	Trimmed-mean Inflation	Output Gap	Change in Nominal Interest Rate
Expected - next 4 quarters	1.51	1.29	0.70	0.09
Expected - next quarter	1.51	1.28	0.72	0.10
Current quarter	1.50	1.26	0.76	0.09
Average past year	1.48	1.22	0.83	0.11
Average past 3 years	1.44	1.15	0.92	0.12
Price level target	1.41	1.12	0.92	0.13

MARTIN's residuals the nominal aggregate rules perform poorly relative to the baseline policy rule. Similar to the baseline rule a nominal GDP target, in either growth rates or levels, performs the worst while the nominal consumption target generates the lowest loss among the nominal aggregates.

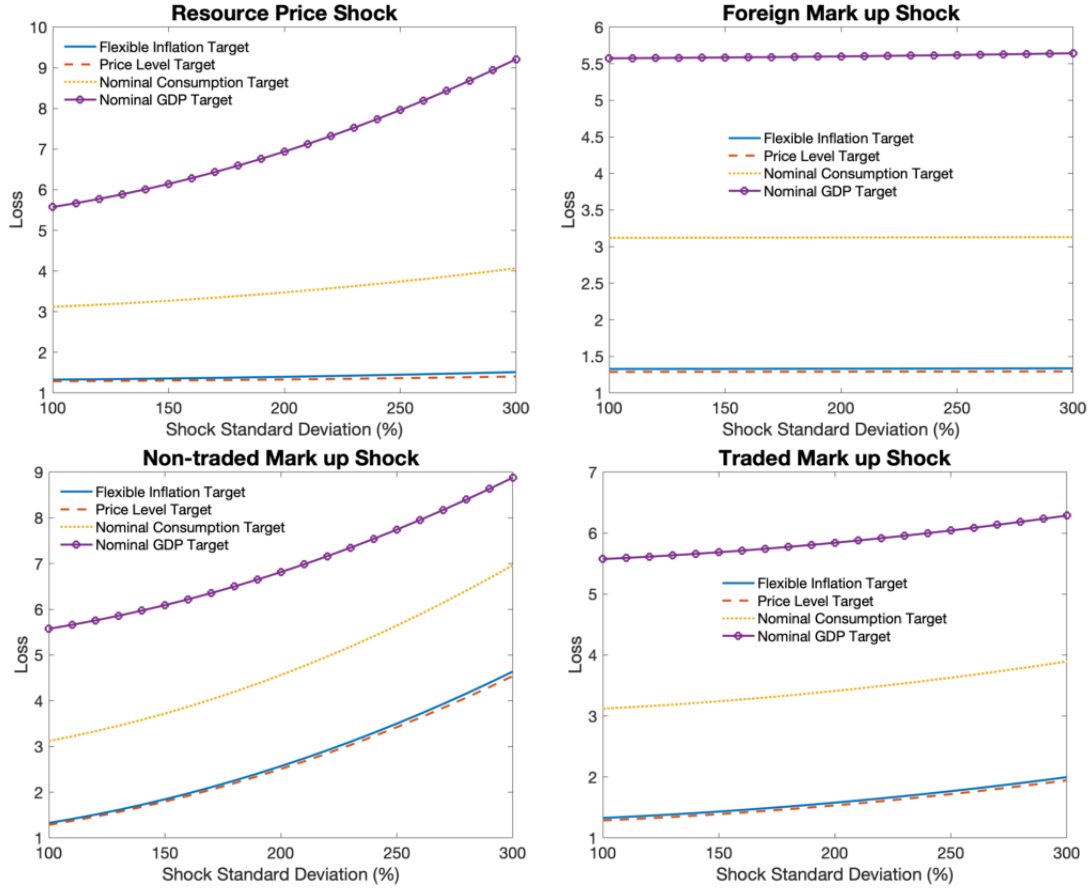
Table 12: Volatilities with Alternative Nominal Aggregate Targets in MARTIN with 10% Higher Residuals

Target	Loss	Trimmed- mean Inflation	Unempl. Gap	NGDP	Change in rates	Nominal Consumption	NDFD	NHCOE
Baseline Policy Rule	1.86	0.66	0.47	9.63	1.43	2.73	4.89	6.44
NGDP Growth	2.83	1.58	0.74	8.63	1.02	2.52	3.50	9.67
NGDP Level	3.66	2.60	0.89	9.72	0.33	3.28	3.80	7.60
Nominal DFD	2.63	1.49	0.73	8.84	0.82	2.51	3.12	9.38
Nominal HCOE	2.50	1.68	0.63	8.43	0.46	2.27	3.39	5.79
Nom. Consumption	2.15	1.06	0.59	9.05	1.00	2.31	3.79	5.97

A policy rule's robustness to higher volatility can also be tested by increasing individual shocks' standard deviation one by one holding everything else constant. We next consider how increases in the volatility of shocks to the resource price, foreign marginal costs, import prices and domestic mark ups affect the welfare loss under a flexible inflation target, a price level target, a nominal consumption target and a nominal GDP target in the MSM.⁸ Unsurprisingly we find that the welfare loss increases monotonically with the standard deviation of the shocks across all policy rules (Figure 4). We also find that the relative ordering of the policy rules does not change as volatility increases. The price level target narrowly dominates the flexible inflation target while the nominal consumption target performs substantively worse, and the nominal GDP target worst of all. Even if the future Australian economy experiences drastically higher climate changed induced volatility it is unlikely to change the relative order of the optimal monetary policy rules.

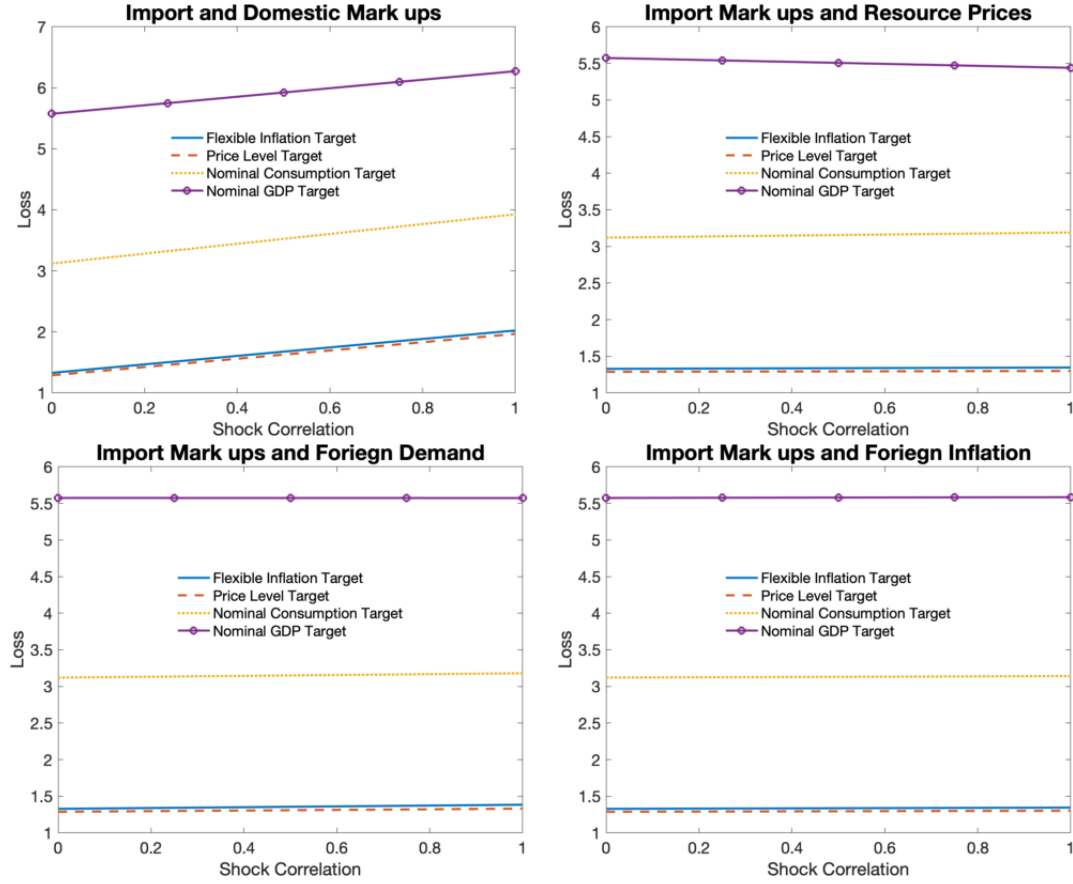
⁸The flexible inflation target and price level target are calibrated to minimise the loss function in the baseline calibration. The nominal consumption and GDP targets are calibrated with a weight of 2 and a weight of 1.5 on inflation to ensure determinacy.

Figure 4: Welfare with Higher Shock Volatility



Finally, we consider the same four policy rules under a counterfactual in which climate change induces a positive correlation in previously uncorrelated exogenous shocks. We look at four counterfactuals in which import price mark ups are correlated with resource prices, non-traded mark ups, foreign demand, and foreign inflation respectively. These examples are informed by McKibbin et al (2020) who consider how climate change might alter the nature of exogenous macroeconomic shocks. Each of these alternative calibrations approximate an environment in which climate driven supply shocks affect both domestic and foreign economies simultaneously (ie increasing both domestic and foreign mark ups) or different aspects of the global economy at the same time (ie increasing demand for raw commodities as well as foreign mark ups). Increasing the correlation between these exogenous shocks from the baseline assumption of zero does not change the ordering of the four monetary policy rules. NGDP targeting performs slightly better when import mark ups and resources prices are positively correlated, but the improvement is small (Figure 5). It remains strictly dominated by a nominal consumption target, which is in turn strictly dominated by a flexible inflation target or price level target.

Figure 5: Welfare with Correlated Exogenous Shocks



6 Conclusion

We find that when attempting to minimise the deviations of inflation and unemployment from their respective targets it is hard to improve upon a policy rule which sets the cash rate in response to the aforementioned deviations. This analysis suggests that the RBA's dual mandate is best achieved by targeting a combination of the unemployment gap and a price level target. Targeting a nominal aggregate variable which combines both a real measure of activity and a measure of price growth in a single variable is not optimal. Of the alternatives considered a nominal consumption growth rate target seems most promising. However when nominal consumption is targeted to the exclusion of other variables welfare is still significantly lower than the baseline policy. Importantly, even if a price level target is considered too radical a change from the status quo, much of its benefit can be achieved by a policy of targeting the average inflation over the past couple of years.

One important limitation of both models used in this analysis is that they largely abstract from non-

linearities in the economy such as the effective lower bound and downwardly rigid nominal wages. The effective lower bound was a key motivation behind the Federal Reserve’s adoption of asymmetric flexible average inflation targeting as it increases inflation expectations when nominal interest rates cannot be lowered, thus decreasing the real interest rate Clarida (2022). A key advantage of price level target or average inflation target is that it increases the expected rate of inflation when monetary policy is constrained by the effective lower bound, helping to lower real interest rates. As such this linear analysis represents the lower bound for the potential gains from adopting an average inflation target or price level target when nominal interest rates are constrained by the effective lower bound. Though it should be noted that Dorich, Mendes and Zhang (2021) find that the assumption of an effective lower bound does not change the overall qualitative pattern of the policy rules they consider. However a price level target, which requires an equally sized disflation to offset any unforeseen overshoot of the inflation target, may be more costly if nominal rigidities are downwardly rigid.

This analysis suggests that the current approach of implementing the RBA’s dual mandate in the form of a policy rule that places responds to both trimmed-mean inflation and the unemployment rate remains close to optimal. However the status quo of letting bygones be bygones could be improved by adopted a policy of either averaging inflation over a short horizon or adopting a price level target.

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Appendix

We use a version of MARTIN that closely follows that published by the RBA in Ballantyne et al. (2020). However, there are some series that had to be substituted because public versions of the data do not exist. This version of the MARTIN was developed by Stephan (2019) and is available at:

<https://github.com/MacroDave/MARTIN>.

While the RBA model has data on net asset transfer between the public and private sectors, this is not publicly available which results in minor differences in non-mining business investment and government investment. Similarly, instead of calculating data on world GDP or weighting the overseas variables by major trading partner, we simply use US data for these series. This is primarily due to the lack of quarterly data from China. Finally, we use a fixed time trend for the export sub-sector equations which cover mining, manufacturing and the service sectors.

The version of the MSM we use is closely derived from Rees, Hall & Smith (2016). The only change, reflecting further model development at the RBA, is to add nominal wage rigidity in the non-tradable, non-resource tradable and resource sector. This requires the assumption that workers have heterogeneous skill which gives them market power to negotiate their nominal wage subject to firms' demand, Rotemberg wage adjustment costs and partial wage indexation. This generates the wage Phillips curve

$$\pi_{t,j}^w = \frac{\xi_{w,j}}{1 + \beta\xi_{w,j}}\pi_{t-1,j}^w + \frac{\beta}{1 + \beta\xi_{w,j}}E_t\{\pi_{t+1,j}^w\} - \frac{\kappa_j}{1 + \beta\xi_{w,j}}(\tilde{w}_{t,j}) + \epsilon_{t,j}^w$$

where wage inflation, $\pi_{t,j}^w$, is determined by past and expected wage inflation, the real wage gap $\tilde{w}_{t,j}$ and wage mark up shocks $\epsilon_{t,j}^w$. Data on nominal wage growth for workers in the non-tradable, tradable and resource sector are included in the estimation process.